

Learning objectives

In this lesson you will learn to:

- Explain how data is protected by encryption
- Describe how backup and recovery procedures protect against data loss
- Explain how access control helps to protect systems and data.

For more information and additional learner activities see Topic 5.3 of the student book.

Protecting data

In the last lesson, you learned how Acceptable Use Policies (AUPs) help protect systems from social engineering attacks.

The human aspect of computer systems is just one part of the story. Much of the operation of computer systems in the modern world involves the automated storage, manipulation and transfer of huge quantities of data.

In this lesson you will learn how this data can be protected.

Encryption

When sensitive data is stored or transmitted it can be accessed by hackers. They can do this either by breaking into a system or by intercepting a data transmission.

To protect data in these situations, it should be encrypted.

Encryption is the process of converting data into an illegible, scrambled format so that it cannot be understood by unauthorised parties.

Data may still be accessed and read by a hacker, but because it is scrambled, it will not be understandable.

Backup and recovery

Backing up data involves taking a copy of the data and storing it in a different location – for example, in the ‘cloud’.

Backing up is often automated and should be done frequently – daily or hourly for critical data and systems.

Recovery is the process of using a back-up copy of data to restore a database or system to the correct state when a failure occurs.

Access control

Users have different levels of access to systems (this is much like the read/write/delete permissions you learned about in the lesson on file management).

Users should only be given access to the software, data and hardware that they require (known as the principle of least privilege).

This helps to ensure that users cannot accidentally (or maliciously) damage systems or data.

Physical security

Another way that systems and data may become lost or physically harmed is through theft, mistreatment or damage.

Common physical security measures include fire detection and suppression systems, CCTV, locks, biometrics, and ensuring that hardware is in a safe location.

Wrap up: you have learned how to...

- ✓ Explain how data is protected by encryption.
 - Encryption scrambles data so that it cannot be understood if it is intercepted.
- ✓ Describe how backup and recovery procedures protect against data loss.
 - Backup is effective only if completed regularly.
 - Incremental and full backups can be used to create copies of data that can be restored later if the original is lost or damaged.
- ✓ Explain how access control helps to protect systems and data.
 - Applying the principle of least privilege means that users only have access to the data and systems they need and nothing more, so limiting the harm they can do.